

ON THE IMPORTANCE OF ATTACK INITIALIZATION

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ABSTRACT

Targeted clean-label data poisoning attacks aim to force a model to misclassify a specific target instance by injecting imperceptibly perturbed training examples. While prior research primarily focuses on designing advanced perturbation optimizers, this paper demonstrates that attack success depends critically on the initialization phase—specifically, the combinatorial selection of clean base examples to perturb. We introduce GRAFT (Gradient-based Ranking and Alignment for Poisoning Targeting), a target-conditioned base-selection algorithm that ranks candidate base examples prior to optimization by evaluating how strongly their induced parameter updates reduce the adversarial loss at the target sample. Through extensive evaluations across multiple architectures (ConvNet, ResNet-20, VGG-13) and datasets (CIFAR-10, CIFAR-100, SVHN, and Tiny ImageNet), we show that GRAFT significantly amplifies attack success rates over random selection while preserving clean test accuracy. Furthermore, our approach demonstrates robust transferability across architectures and maintains high efficacy against standard victim-side defenses and data augmentations.

1 INTRODUCTION

Deep neural networks are notoriously vulnerable to data poisoning attacks, where malicious entities inject subtle, clean-looking perturbations into the training set to manipulate model behavior at inference time. In the context of targeted clean-label poisoning, the adversary seeks to cause a specific test target to be misclassified while keeping the true labels of the training data correct and human-perceptible. Existing literature has predominantly concentrated on refining the downstream perturbation optimization process, utilizing techniques such as Bullseye Polytope, Witches’ Brew, and SAPA to generate the poisoned inputs.



Despite these advances, a critical blind spot has persisted in existing threat models: the selection of which clean base training examples to perturb in the first place. Traditionally, base examples are chosen at random, an approach that overlooks how the choice of initial data points profoundly influences the downstream optimization landscape and final attack success. Evaluating defenses solely against randomly chosen base examples significantly underestimates the true vulnerabilities of victim models.

To bridge this gap, we establish that initialization is a primary driver of clean-label poisoning efficacy. We propose GRAFT (Gradient-based Ranking and Alignment for Poisoning Targeting), an algorithm-agnostic, target-conditioned base-selection method. GRAFT leverages an ensemble of surrogate models to rank candidate base training examples by computing their gradient alignment with the adversarial target. By prioritizing bases that yield strong directional parameter updates toward reducing the target loss, GRAFT consistently outperforms random selection. Our comprehensive evaluations confirm that strategically initialized attacks effectively pierce standard victim-side data augmentations, bypass active defenses like EPIC and FRIENDS, and successfully scale across diverse multi-class datasets and neural architectures without degrading overall clean test accuracy.

2 RELATED WORK

Targeted Clean-label Data Poisoning. Data poisoning attacks modify the training set to change the behavior of a learned model. We study *targeted clean-label poisoning*, where an attacker makes small, visually imperceptible changes to a limited number of correctly labeled training examples. The goal is to make an unmodified target x_t be classified as a chosen adversarial class y_{adv} without noticeably reducing overall test accuracy Shafahi et al. (2018); Zhu et al. (2019); Aghakhani et al. (2021); Huang et al. (2020); Geiping et al. (2020). Unlike backdoor poisoning, this setting does not learn a shared trigger that must be added to inputs at test time Gu et al. (2017); Turner et al. (2019); Saha et al. (2020); Souri et al. (2022). Early clean-label attacks mainly used the positions of examples in feature space. Feature Collision moves a perturbed base image toward the target representation Shafahi et al. (2018), while Convex Polytope jointly constructs several poisons so that their representations surround the target Zhu et al. (2019). Bullseye Polytope further encourages the target to lie near the center of the poison representations, improving scalability and transferability Aghakhani et al. (2021). These attacks were less effective when the victim model was trained from scratch Schwarzschild et al. (2021), which led to methods that account more directly for the training process. MetaPoison uses bilevel optimization and differentiates through a shortened victim-training process Huang et al. (2020). Witches’ Brew avoids this step by matching the parameter gradients produced by the poisons and the adversarial target Geiping et al. (2020). More recently, SAPA improves reliability across different victim-training runs by using a sharpness-aware poisoning objective He et al. (2024).

Base Selection in Adversarial Attacks. Several widely used targeted poisoning attacks choose clean bases without an explicit selection objective. For example, MetaPoison and BP construct poisons from the first few examples of the poison class, despite evidence that training examples can differ substantially in their effect on model training Koh & Liang (2017). Sample selection has been studied more explicitly in backdoor poisoning, where methods prioritize examples using gradient norms, forgetting events, sample difficulty, or feature-space abnormality Souri et al. (2022); Xia et al. (2022); Gao et al. (2023); Nguyen et al. (2025). Such selection can improve attack efficiency by identifying examples that more effectively reinforce a shared trigger-to-label association. For instance, difficult examples may provide weaker semantic cues and encourage reliance on the injected trigger for prediction Gao et al. (2023); Nguyen et al. (2025). Related work has also exploited variation in poison importance for defense by identifying influential poisons and reducing their contribution during training Yang et al. (2022). Triggerless targeted poisoning differs fundamentally because there is no shared trigger and each selected base is individually perturbed. A difficult or high-gradient example may strongly affect training while remaining unrelated to the target being attacked. Existing backdoor selection criteria provide useful heuristic baselines, but they do not exploit this target-specific relationship. Among methods that address poisoning initialization more explicitly, the closest prior work is Guided Diffusion Poisoning (GDP) Souri et al. (2024), which synthesizes new base images with a diffusion model before applying a downstream poisoning objective. In contrast, we study target-conditioned selection of existing clean training examples before any poison perturbations are optimized. Because base selection is separated from perturbation construction, it is algorithm-agnostic and can be paired with different downstream clean-label poisoning methods without changing their objectives.

3 METHOD

3.1 DATA POISONING AS A COMBINATORIAL BILEVEL OPTIMIZATION

Consider a clean training dataset $\mathcal{D}_c = \{(x_i, y_i)\}_{i=1}^n$ consisting of n examples from C classes. Let x_t be a clean, unmodified target example with the ground-truth label y_t . The attacker’s objective is to manipulate the victim model’s training data such that, after training, the model misclassifies x_t as a chosen adversarial target class y_{adv} , where $y_{\text{adv}} \neq y_t$. The attacker begins by selecting m training examples labeled as y_{adv} to serve as clean *base samples*, denoted by $\mathcal{X}_c$. Let Δ denote the collection of perturbations applied to the inputs in $\mathcal{X}_c$. With a slight abuse of notation, we let $\mathcal{X}_c + \Delta$ denote the resulting poisoning set, where the addition modifies only the inputs while preserving their labels. To make the attack visually imperceptible, every perturbation in Δ is constrained by an ℓ_∞ -norm

budget ϵ_∞ . The poisoning set replaces its clean base samples, yielding the poisoned training dataset

$$\mathcal{D}_p = (\mathcal{D}_c \setminus \mathcal{X}_c) \cup (\mathcal{X}_c + \Delta).$$

The optimal perturbations depend on the selected base samples because each selection induces a different feasible neighborhood under the same perturbation budget. Consequently, different selections can produce substantially different poisoning sets. The attacker must therefore jointly optimize the combinatorial selection of $\mathcal{X}_c$ and the corresponding perturbations Δ . Over the admissible base-sample selections and perturbations described above, the attack can be formulated as

$$(\mathcal{X}_c^*, \Delta^*) \in \arg \min_{\mathcal{X}_c, \Delta} \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta^*) \quad \text{s.t.} \quad \theta^* \in \arg \min_{\theta} \mathcal{L}_{\text{train}}(\mathcal{D}_p; \theta). \quad (1)$$

Here, $\mathcal{L}_{\text{adv}}$ denotes the adversarial loss, which is minimized when the victim model assigns y_{adv} to x_t , while $\mathcal{L}_{\text{train}}$ denotes the empirical loss used to train the victim model. Optimizing $\mathcal{X}_c$ constitutes the combinatorial component of the outer problem, while optimizing Δ constructs the corresponding poisoned samples. The dependence of the outer objective on the solution of the inner training problem gives the formulation its bilevel structure. The solution yields the optimal base-sample selection $\mathcal{X}_c^*$ and perturbations Δ^* , with $\mathcal{X}_c^* + \Delta^*$ forming the optimal poisoning set.

3.2 STRATEGY FOR SELECTING EFFECTIVE BASE SAMPLES

Equation (1) defines the optimal attack, but solving it directly is computationally intractable. Determining the attack objective for a given subset $\mathcal{X}_c$ requires optimizing its perturbations and solving the corresponding bilevel problem, which cannot be repeated over all possible subsets of clean training samples at practical dataset scales. We therefore begin by choosing clean base samples using a computationally efficient criterion that can be evaluated prior to poison optimization and apply any existing perturbation-based poisoning method to optimize Δ for the selected samples. This selection is motivated by the fact that base samples can differ in how strongly they affect the target prediction after being perturbed Yang et al. (2022). Each clean training sample with label y_{adv} is eligible to serve as a base sample, and we refer to these samples as candidates. To assess the effectiveness of a candidate for the attack, we examine whether a small parameter update induced by its loss would locally reduce $\mathcal{L}_{\text{adv}}$ at x_t . We quantify this effect by comparing the parameter gradient of the candidate loss against the adversarial loss at the target sample. For each candidate x_i , define

$$g_i = \nabla_{\theta} \ell(x_i, y_{\text{adv}}; \theta), \quad g_t = \nabla_{\theta} \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta), \quad (2)$$

where ℓ denotes the per-example loss. To understand how a candidate affects the attack objective, consider a small parameter update induced by its gradient $\theta' = \theta - \eta g_i$, where $\eta > 0$. Now, using first-order Taylor expansion and expanding the adversarial loss at x_t around θ gives

$$\mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta') = \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta) - \langle \nabla_{\theta} \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta), \theta - \theta' \rangle + \mathcal{O}(\eta^2).$$

The second term on the R.H.S. of the above equation evaluates to $\eta \langle g_i, g_t \rangle$. Therefore, in order to reduce the adversarial loss at x_t , we need a positive and preferably large $\langle g_i, g_t \rangle$. To expose the structure of this alignment, we write the model as $z_{\theta}(x) = W h_{\phi}(x)$, and $\theta = (W, \phi)$, where $h_{\phi}(x) \in \mathbb{R}^d$ is the representation before the final linear classifier $W \in \mathbb{R}^{C \times d}$. Define

$$r_i = \nabla_{z_{\theta}(x_i)} \ell(x_i, y_{\text{adv}}; \theta), \quad r_t = \nabla_{z_{\theta}(x_t)} \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta).$$

The full parameter-gradient alignment can then be written as

$$\langle g_i, g_t \rangle = r_i^{\top} \left[(h_i^{\top} h_t) I + W \frac{\partial h_{\phi}(x_i)}{\partial \phi} \frac{\partial h_{\phi}(x_t)}{\partial \phi}^{\top} W^{\top} \right] r_t. \quad (3)$$

Further details and derivations leading to the above equation are provided in Appendix C. Equation (3) shows that when the candidate residual r_i is close to zero, the gradient alignment vanishes, regardless of the remaining terms. Under softmax cross-entropy, $r_i = p_{\theta}(x_i) - e_{\text{adv}}$. Consequently, samples that the model already predicts as y_{adv} with high confidence induce only a small gradient toward the attack objective and are therefore ranked lower. Beyond the residual term, the alignment also depends on the candidate–target feature similarity $h_i^{\top} h_t$ and on how similarly their representations vary with respect to the model parameters.

Motivated by the decomposition above, we propose GRAFT, a target-conditioned base-selection method that converts the alignment components in Equation (3) into a practical score for ranking clean candidates. A central challenge in constructing this score is that the residual r_i , candidate–target feature similarity, and remaining alignment terms are model-dependent and may differ between the attacker’s surrogate models and the unknown victim model. We therefore reduce sensitivity to this surrogate–victim mismatch by ranking candidates using multiple independently trained surrogate models and selects the top- m samples that consistently receive high rankings across surrogates. More formally, we train K independent surrogate models and aggregate the aforementioned quantities across them, reducing the sensitivity of the ranking to any particular surrogate model. Specifically, for each candidate x_i and surrogate model k , we use M_i as a proxy for the residual magnitude to penalize samples that the model classifies with high confidence as their true label, resulting in a small $\langle g_i, g_t \rangle$. We capture the remaining alignment factors through the feature similarity R_i and the parameter-gradient interaction A_i . We average these quantities across the K surrogates, standardize them over the candidate pool, and combine them into the final ranking score s_i . The hyperparameter β controls the contribution of A_i ; setting $\beta = 0$ avoids the costly per-sample gradient computation. Empirically, including A_i provides only marginal gains over using M_i and R_i alone. Our ablations suggest that this is partly explained by the strong correlation between $h_i^\top h_t$ and the corresponding parameter-gradient interaction $\frac{\partial h_\phi(x_i)}{\partial \phi} (\frac{\partial h_\phi(x_t)}{\partial \phi})^\top$ Jacot et al. (2018). Finally, we select the m candidates with the highest scores for downstream poison optimization.

Algorithm 1 GRAFT

Require: Clean data $\mathcal{D}_c$, target x_t , adversarial label y_{adv} , budget m , K surrogates, $\beta \in \{0, 1\}$

- 1: **Begin**
- 2: Train K independent surrogates on $\mathcal{D}_c$
- 3: **for** each candidate x_i with $y_i = y_{\text{adv}}$ **do**
- 4: **for** $k = 1, \dots, K$ **do**
- 5: $M_i^{(k)} \leftarrow z_{y_{\text{adv}}}^{(k)}(x_i) - \max_{c \neq y_{\text{adv}}} z_c^{(k)}(x_i)$
- 6: $R_i^{(k)} \leftarrow \langle h_i^{(k)}, h_t^{(k)} \rangle$
- 7: $A_i^{(k)} \leftarrow \langle \nabla_{\phi} \ell_i^{(k)}, \nabla_{\phi} \mathcal{L}_{\text{adv}, t}^{(k)} \rangle$
- 8: **end for**
- 9: Average $M_i^{(k)}$, $R_i^{(k)}$, and $A_i^{(k)}$ over the K
- 10: **end for**
- 11: Normalize each averaged quantity across candidates, yielding $\tilde{M}_i$, $\tilde{R}_i$, and $\tilde{A}_i$
- 12: $s_i \leftarrow \tilde{R}_i - \tilde{M}_i + \beta \tilde{A}_i$
- 13: **Return** the m candidates with the largest s_i

4 EXPERIMENTS

We evaluate whether GRAFT, our target-conditioned base-selection method, strengthens clean-label poisoning attacks against CIFAR-10 classifiers trained from scratch and whether the resulting gains generalize across victim architectures, data-augmentation settings, victim-side defenses, and different perturbation-optimization objectives. For each attack instance, the attacker has access to the clean training set $\mathcal{D}_c$ and a target pair (x_t, y_{adv}) , where x_t is an unmodified CIFAR-10 test example and every training example labeled y_{adv} is an eligible base candidate. Using Algorithm 1 and $K = 20$ independently trained surrogate models, we rank the eligible candidates and select m of them to form $\mathcal{X}_c$. We express the poisoning budget as $\rho = m/n$, where n is the size of the full clean training set. The selected bases are passed to Bullseye Polytope (BP) Aghakhani et al. (2021), Witches’ Brew Gradient Matching (GM) Geiping et al. (2020), and Sharpness-Aware Attack (SAPA) He et al. (2024), each of which retains its original perturbation-optimization objective. The downstream optimizer constructs label-preserving perturbations Δ such that every $\delta_i \in \Delta$ satisfies $\|\delta_i\|_\infty \leq \epsilon_\infty = 8/255$. The resulting samples $\mathcal{X}_c + \Delta$ replace their clean counterparts in $\mathcal{D}_c$ to form $\mathcal{D}_p$, while all other training examples remain unchanged. We train ConvNet, ResNet-20, and VGG-13 victims from scratch on $\mathcal{D}_p$ and evaluate every experimental configuration using 10 targets and six independent victim-training runs per target. We compare GRAFT with RAND, which draws a fresh set of eligible bases for each target and downstream poison optimizer. In each comparison, only the base-selection method differs, and all other parameters and protocols stay the same. Attack success rate (ASR) is the fraction of victim-training runs whose trained model assigns x_t to the specified y_{adv} , with every other prediction counted as a failure. Clean test accuracy (CTA) measures predictive performance on the clean test set. We omit data augmentation by default and examine its effect in a separate ablation. We also provide additional implementation details in Appendix D.

Table 1: Attack success rate for targeted clean-label poisoning across poison ratios ρ . We compare RAND, GRAFT, and GRAFT⁺. Arrows indicate $y_t \rightarrow y_{adv}$. Entries report ASR (%), $\uparrow$.

		dog $\rightarrow$ bird									frog $\rightarrow$ airplane								
ρ ($\times 10^{-3}$)		BP			GM			SAPA			BP			GM			SAPA		
		RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺
ConvNet	1	0.0	28.3	41.7	0.0	11.1	45.0	0.0	21.7	31.2	0.0	5.0	3.3	0.0	2.5	5.0	0.0	3.3	2.5
	2	1.7	65.0	70.0	0.0	28.3	53.3	0.0	26.7	45.8	0.0	8.3	10.0	0.0	3.3	5.0	0.0	8.3	10.0
	5	13.3	88.3	95.0	1.7	72.9	61.7	6.7	48.3	71.2	0.0	26.7	26.7	0.0	4.2	15.0	0.0	5.0	11.7
	10	35.0	96.7	95.0	21.7	85.4	87.9	23.3	81.7	88.8	1.7	36.7	45.0	3.3	20.0	16.7	1.7	6.7	21.7
	20	73.3	98.3	93.3	43.3	95.0	100	50.0	100	100	10.0	55.0	55.0	3.3	35.0	47.5	8.3	25.0	35.0
40	91.7	98.3	100	83.3	100	100	76.7	100	100	100	30.0	66.7	63.3	16.7	52.5	55.0	25.0	60.0	67.5
ResNet20	1	11.7	17.5	12.5	15.0	35.0	40.0	18.3	42.5	37.5	8.3	2.5	12.5	1.7	22.5	22.5	1.7	27.5	32.5
	2	16.7	25.0	37.5	31.7	55.8	60.0	21.7	52.5	67.5	8.3	6.7	5.0	0.0	36.7	26.7	5.0	20.0	35.0
	5	30.0	46.7	27.5	43.3	76.7	82.5	55.0	77.5	75.0	18.3	24.1	25.0	8.3	60.0	61.7	3.3	50.0	63.3
	10	43.3	55.0	50.0	56.7	90.8	85.0	65.0	85.0	85.0	16.7	40.0	22.5	26.7	74.6	68.5	21.7	66.2	75.0
	20	55.0	33.3	33.3	75.0	93.8	100	95.0	100	100	30.0	41.7	41.7	45.0	89.6	85.4	40.0	83.3	85.7
40	48.3	20.0	18.8	98.3	100	100	98.3	100	100	100	33.3	28.3	25.0	70.0	100	97.6	81.7	94.4	94.4
VGG13	1	21.7	27.5	30.0	13.3	52.5	70.0	11.7	52.5	87.5	1.7	7.5	7.5	0.0	40.0	40.0	0.0	37.5	45.0
	2	41.7	36.7	37.5	21.7	56.2	77.5	26.7	82.5	87.5	1.7	3.3	12.5	5.0	46.7	56.7	10.0	47.5	51.7
	5	45.0	51.7	55.0	55.0	95.0	100	60.0	95.0	95.0	3.3	8.3	12.5	16.7	66.7	80.0	15.0	65.8	70.0
	10	41.7	53.3	55.0	76.7	95.8	100	88.3	100	100	13.3	30.0	20.0	28.3	83.3	80.0	20.0	84.6	80.8
	20	61.7	64.6	56.2	96.7	100	100	100	100	100	10.0	16.7	31.2	56.7	87.5	85.4	53.3	85.4	91.7
40	75.0	41.7	60.4	98.3	100	100	100	100	100	100	16.7	20.0	10.4	78.3	89.6	87.5	85.0	87.5	100

4.1 MAIN RESULTS

Table 1 reports ASR for RAND, GRAFT with $\beta = 0$, and GRAFT with $\beta = 1$, which we denote as GRAFT⁺. Over the 108 configurations, RAND base selection yields a mean ASR of 31.2%. GRAFT increases the mean ASR to 54.4%, and GRAFT⁺ further raises it to 57.7%. Relative to RAND, these correspond to improvements of 23.2 and 26.5 percentage. GRAFT outperforms RAND in 99 configurations, whereas GRAFT⁺ does so in 97. Both methods increase the average ASR across all victim architectures and target directions. The improvements are particularly strong at smaller poisoning budgets. For $\rho \leq 5 \times 10^{-3}$, the average ASR for RAND is 12.4% and rises to 37.2% for GRAFT and 43.4% for GRAFT⁺. Both methods also achieve higher ASR than RAND in 51 of the 54 settings. As ρ increases, RAND becomes more likely to include influential samples for attack initialization, and several configurations approach 100% ASR, leaving less room for further improvement using GRAFT. However, for $\rho \geq 0.02$, GRAFT and GRAFT⁺ still achieve average gains of 15.3 and 16.9 percentage over RAND. Comparing GRAFT with GRAFT⁺, the latter increases the average ASR by an additional 3.3 percentage points relative to the former. Moreover, base selection with GRAFT has a different effect on BP at large poisoning budgets and results in a decline in ASR. One possible explanation is that BP optimizes the perturbations so that the average representation of the perturbed bases remains close to the target representation. At large budgets, GRAFT may select a set whose average clean representation is already close to the target before any perturbation is added. BP therefore starts with a small objective value and produces weak perturbations. Because the resulting samples remain close to their original versions, they provide a weaker poisoning signal during victim training, which may explain the decrease in ASR at large budgets using BP.

We next assess whether the ASR improvements achieved by GRAFT are accompanied by a loss in CTA. Figure 1 shows that for the dog $\rightarrow$ bird setting, the CTA of GRAFT⁺ remains close to and is sometimes slightly higher than the CTA achieved by RAND across different configurations, despite the large improvement in ASR achieved by GRAFT⁺. This behavior is consistent with GRAFT favoring lower-margin bases near the decision boundary instead of modifying examples in high-confidence classification regions. We also observe a small decrease in CTA for both selection methods at larger poisoning budgets, as expected when more training examples are modified. Complete CTA results for all Table 1 configurations, with additional statistical tests, are in Appendix E.

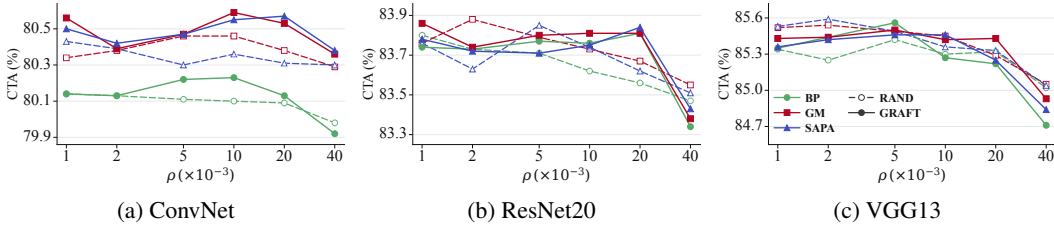


Figure 1: CTA (%), $\uparrow$) under varying perturbation budget ρ for the dog-bird class pair.

Table 2: Cross-architecture transfer on dog→bird. Rows show surrogate architecture S ; column show victim architecture and K . Cells report ASR (RAND/GRAFT); shading marks $S = V$.

Attack	ρ ($\times 10^{-3}$)	Selector S	ConvNet			ResNet20			VGG13		
			$K = 1$	$K = 3$	$K = 20$	$K = 1$	$K = 3$	$K = 20$	$K = 1$	$K = 3$	$K = 20$
BP	2	ConvNet	0.0/55.0	0.0/75.0	1.7/65.0	20.0/20.0	10.0/25.0	5.0/35.0	10.0/55.0	25.0/55.0	10.0/70.0
		ResNet20	0.0/20.0	0.0/30.0	0.0/45.0	15.0/35.0	5.0/30.0	16.7/25.0	15.0/40.0	10.0/30.0	10.0/50.0
		VGG13	5.0/5.0	0.0/25.0	5.0/25.0	5.0/20.0	25.0/30.0	15.0/40.0	10.0/55.0	20.0/45.0	41.7/36.7
	5	ConvNet	10.0/85.0	20.0/90.0	15.0/88.3	15.0/40.0	15.0/55.0	15.0/65.0	25.0/70.0	30.0/75.0	40.0/90.0
		ResNet20	20.0/50.0	15.0/50.0	5.0/70.0	10.0/40.0	25.0/40.0	35.0/46.7	30.0/60.0	25.0/75.0	50.0/65.0
		VGG13	15.0/45.0	20.0/60.0	15.0/55.0	25.0/40.0	10.0/45.0	35.0/30.0	40.0/35.0	25.0/30.0	35.0/51.7
GM	2	ConvNet	0.0/30.0	0.0/30.0	0.0/28.3	5.0/55.0	25.0/65.0	30.0/55.0	10.0/85.0	5.0/95.0	20.0/90.0
		ResNet20	0.0/5.0	0.0/15.0	0.0/15.0	30.0/55.0	25.0/70.0	31.7/55.8	15.0/65.0	10.0/80.0	15.0/75.0
		VGG13	5.0/5.0	0.0/5.0	0.0/10.0	20.0/50.0	25.0/35.0	15.0/35.0	10.0/30.0	15.0/55.0	21.7/56.2
	5	ConvNet	0.0/45.0	0.0/35.0	0.0/72.9	45.0/60.0	35.0/55.0	45.0/85.0	50.0/100	65.0/85.0	65.0/100
		ResNet20	0.0/0.0	5.0/25.0	0.0/30.0	40.0/80.0	30.0/75.0	35.0/76.7	50.0/85.0	55.0/100	60.0/100
		VGG13	5.0/20.0	10.0/30.0	0.0/25.0	35.0/60.0	25.0/75.0	55.0/95.0	65.0/80.0	50.0/95.0	35.0/95.0
SAPA	2	ConvNet	0.0/25.0	0.0/25.0	0.0/26.7	25.0/60.0	15.0/50.0	15.0/65.0	10.0/85.0	15.0/90.0	30.0/100
		ResNet-20	0.0/5.0	0.0/0.0	5.0/25.0	35.0/50.0	5.0/35.0	21.7/52.5	0.0/80.0	5.0/60.0	10.0/75.0
		VGG-13	0.0/0.0	0.0/0.0	0.0/10.0	20.0/35.0	15.0/55.0	20.0/65.0	10.0/40.0	10.0/50.0	26.7/82.5
	5	ConvNet	0.0/40.0	0.0/40.0	6.7/48.3	25.0/70.0	35.0/60.0	20.0/70.0	70.0/100	80.0/100	35.0/95.0
		ResNet-20	5.0/10.0	5.0/20.0	0.0/30.0	40.0/65.0	45.0/80.0	55.0/77.5	80.0/90.0	65.0/90.0	65.0/100
		VGG-13	5.0/30.0	5.0/35.0	5.0/30.0	35.0/65.0	50.0/75.0	30.0/90.0	75.0/80.0	40.0/80.0	60.0/95.0
Average for $S = V$:			$K = 1$	21.7 / 51.4 (+29.7)		$K = 3$	17.5 / 54.4 (+36.9)		$K = 20$	24.4 / 60.0 (+35.7)	
Average for $S \neq V$:			$K = 1$	19.4 / 46.8 (+27.4)		$K = 3$	20.4 / 51.5 (+31.1)		$K = 20$	20.7 / 58.6 (+37.9)	

4.2 ABLATION ON CROSS-ARCHITECTURE TRANSFER AND SURROGATE ENSEMBLE SIZE

In a realistic threat model, an attacker rarely knows the exact architecture of the victim model. This ablation examines whether the base-selection advantages of GRAFT persist when the surrogate models (S) used to score candidates differ from the victim model (V). The analysis additionally explores how scaling the surrogate ensemble size (K) impacts the attack success under both matched ($S = V$) and transfer ($S \neq V$) conditions. To strictly isolate the influence of the selection architecture, any further surrogate models needed by the downstream perturbation optimizer are set to align with the victim. Results across these configurations are summarized in Table 2. Focusing on the default setting of $K = 20$, the average ASR of GRAFT decreases only slightly from 60.0% when $S = V$ to 58.6% when $S \neq V$. GRAFT also maintains a notable advantage over the RAND baseline under architecture transfer, improving the ASR by 37.9% compared to a 35.7% gain when the architectures match. Furthermore, the 1.4% performance drop in GRAFT is notably smaller than the 3.7% fluctuation between the corresponding RAND averages. Since random selection is entirely independent of S , this variation in RAND originates from underlying differences between the evaluation partitions rather than any actual effect of the selector architecture. These results show that GRAFT yields candidate rankings that generalize well to unseen architectures used by the victim.

Beyond the effects of cross-architecture transfer, utilizing a larger number of surrogates for base selection consistently improves performance in both matched and transfer settings. Under the matched condition ($S = V$), the average ASR of GRAFT climbs from 51.4% with a single surrogate to 54.4% with three, and reaches 60.0% with an ensemble of twenty. Similarly, the transfer setting ($S \neq V$) sees averages increase from 46.8% at $K = 1$ to 51.5% for $K = 3$ and 58.6% at $K = 20$. The absolute gain from $K = 1$ to $K = 20$ is more pronounced under cross-architecture transfer, with an 11.8% gain in the transfer setting compared to an 8.7% gain in the matched setting. Consequently, as the ensemble size grows, the performance gap between matched and cross-architecture selection narrows from 4.6% at $K = 1$ to 2.9% at $K = 3$, and further to 1.4 percentage at $K = 20$. While this improvement is not strictly monotonic in every individual configuration, the overall pattern is consistent. Across the full set of comparisons, the $K = 20$ ensemble matches or surpasses the $K = 1$ baseline in 46 out of 54 cases. Overall, these results confirm that GRAFT remains highly effective even when the attacker lacks exact knowledge of the victim architecture. Moreover, scaling the surrogate ensemble further strengthens this transferability. This behavior aligns with the intuition that aggregating gradients across multiple independently trained models yields a base ranking that is less sensitive to the specific characteristics of any single training trajectory or model architecture.

4.3 ABLATION ON VICTIM-SIDE DEFENSES

Figure 2 compares RAND, GRAFT, and GRAFT⁺ under the defense mechanisms of EPIC Yang et al. (2022) and FRIENDS Liu et al. (2022) across all three victim architectures. These defenses target complementary aspects of poisoned training and test whether the gains from target-conditioned base selection persist beyond the undefended setting. We keep the optimized poisons fixed and modify only victim training using these defense methods. EPIC provides a particularly direct test of GRAFT. While GRAFT prioritizes bases that induce a strong target-aligned training signal, EPIC suppresses examples in low-density regions of gradient space based on the observation that attack success may depend on a limited subset of effective poisons. If the gains from GRAFT arose only from concentrating the attack on a few unusually influential bases, EPIC should preferentially remove this advantage. Instead, both GRAFT and GRAFT⁺ outperform RAND in every evaluated configuration. FRIENDS evaluates a complementary defense mechanism by combining additive noise with optimized perturbations during victim training to alleviate the sharp loss regions induced by poisoned examples. FRIENDS is more suppressive overall, particularly for the deeper victim architectures, yet both GRAFT variants show more robustness to defense and outperform RAND in every evaluated configuration. GRAFT⁺ achieves the highest mean ASR, although it does not uniformly improve upon GRAFT. These findings do not suggest that either defense fails. Instead, they indicate that testing with randomly chosen bases can underestimate the remaining vulnerability of defended victim training.

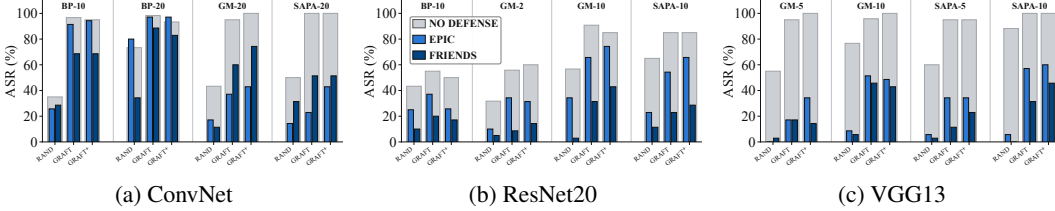


Figure 2: ASR (%), $\uparrow$ under the evaluated defenses across attack configurations and selection methods.

4.4 ABLATION ON VICTIM-SIDE DATA AUGMENTATION

Table 3 evaluates whether the advantage of GRAFT persists when the victim is trained with standard data augmentations. We consider Crop+Flip, RandAugment, and Cutout across multiple attack objectives, poisoning budgets, and victim architectures. These augmentations are applied during victim training, while the surrogate models used for base selection are trained without data augmentation. This setting therefore tests whether the bases selected by GRAFT remain effective when the victim is trained under transformations not used by the selection surrogates. Data augmentation substantially reduces the absolute attack success rate of both selection methods relative to training without augmentation, however, the benefit of target-conditioned base selection remains evident. Across the 33 configurations with augmentation, GRAFT strictly outperforms RAND in 31 and matches it in the remaining two. Averaged over the evaluated attack, budget, and architecture settings, replacing RAND with GRAFT increases ASR from 33.2% to 58.6% under Crop+Flip, from 14.5% to 45.3% under RandAugment, and from 15.0% to 42.3% under Cutout, corresponding to gains of 25.5, 30.8, and 27.3 percentages, respectively. The persistence of GRAFT under augmentation suggests that stronger initialization makes the poisoning signal less sensitive to input transformations. Although victim-side augmentation can disrupt features introduced by the optimized perturbations, GRAFT selected bases show a stronger target-relevant signal across transformed views.

Importantly, these large differences in attack success are not accompanied by a meaningful reduction in clean test accuracy. Across all entries in Table 3, the average CTA difference between GRAFT and RAND is only 0.05%, and the largest absolute difference is below 0.5%. Overall, victim-side augmentation can weaken targeted clean-label poisoning in absolute terms, but it does not remove the advantage provided by stronger attack initialization. Using randomly selected poison bases can make data augmentation appear more effective than it actually is, since stronger base selection with GRAFT reveals that the victim can remain vulnerable to poisoning.

5 CONCLUSION

This paper establishes attack initialization as a critical, previously overlooked dimension of targeted clean-label data poisoning. By shifting focus from the perturbation objective to the strategic selection of base examples, we demonstrate that deep neural networks are significantly more vulnerable to poisoning than previously estimated. To this end, we introduce GRAFT as a target-conditioned selection algorithm prior to perturbation optimization. Extensive evaluations reveal that GRAFT consistently amplifies the attack success of existing optimizers without degrading clean test accuracy. Crucially, these initialization advantages transfer robustly to unseen architectures and enhance attack resilience against standard defenses. Ultimately, our findings reveal a blind spot in current threat models by showing that effective poisoning depends not only on the perturbation itself but also on how the attack is initialized. Recognizing the impact of sample selection on attack success also provides a new direction for evaluating and designing robust defenses against targeted attacks.

Table 3: Victim-side augmentation robustness on dog→bird. Each entry reports ASR(%) with CTA(%) in parentheses. **Bold** denotes GRAFT; normal weight denotes RAND.

		BP ($\rho = 0.002$)	BP ($\rho = 0.01$)	BP ($\rho = 0.02$)	GM ($\rho = 0.02$)
ConvNet	No aug.	5.0 (79.86) / 60.0 (79.94)	50.0 (79.97) / 95.0 (79.91)	85.0 (80.12) / 100 (80.02)	90.0 (80.63) / 95.0 (80.40)
	Crop+Flip	0.0 (85.52) / 5.0 (85.38)	0.0 (85.19) / 20.0 (85.03)	5.0 (85.43) / 30.0 (85.37)	5.0 (86.03) / 45.0 (86.03)
	RandAug.	0.0 (84.19) / 0.0 (84.20)	0.0 (84.27) / 20.0 (83.81)	10.0 (83.80) / 30.0 (83.92)	0.0 (85.66) / 5.0 (85.64)
	Cutout	5.0 (85.43) / 5.0 (85.47)	0.0 (85.42) / 20.0 (85.23)	5.0 (85.58) / 15.0 (85.40)	10.0 (86.12) / 35.0 (85.98)
		GM ($\rho = 0.002$)	GM ($\rho = 0.01$)	GM ($\rho = 0.02$)	SAPA ($\rho = 0.01$)
ResNet20	No aug.	25.0 (83.90) / 60.0 (83.83)	70.0 (83.76) / 90.0 (83.83)	75.0 (83.55) / 85.0 (83.62)	65.0 (83.68) / 90.0 (83.82)
	Crop+Flip	30.0 (89.61) / 60.0 (89.61)	60.0 (89.65) / 65.0 (89.54)	80.0 (89.54) / 95.0 (89.30)	50.0 (89.56) / 80.0 (89.56)
	RandAug.	5.0 (90.39) / 26.3 (90.18)	5.0 (90.36) / 63.2 (90.13)	60.0 (90.07) / 84.2 (89.93)	20.0 (90.31) / 75.0 (90.11)
	Cutout	10.0 (89.99) / 35.0 (89.96)	25.0 (89.91) / 50.0 (89.84)	40.0 (89.91) / 75.0 (89.67)	30.0 (89.88) / 50.0 (89.89)
		GM ($\rho = 0.005$)	GM ($\rho = 0.01$)	SAPA ($\rho = 0.01$)	
VGG13	No aug.	40.0 (85.24) / 95.0 (85.30)	75.0 (85.44) / 100 (85.46)	90.0 (85.22) / 100 (85.30)	
	Crop+Flip	25.0 (90.37) / 60.0 (90.60)	60.0 (90.40) / 90.0 (90.58)	50.0 (90.54) / 95.0 (90.47)	
	RandAug.	10.0 (91.23) / 50.0 (91.23)	20.0 (91.27) / 65.0 (91.34)	30.0 (91.26) / 80.0 (91.16)	
	Cutout	10.0 (90.96) / 25.0 (91.14)	10.0 (91.00) / 70.0 (90.99)	20.0 (91.13) / 85.0 (90.97)	

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A APPENDIX

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B ADDITIONAL RELATED WORK

Several works study clean-label poisoning under threat models related to ours. CLPA Zhao & Lao (2022) examines availability attacks that use correctly labeled samples to degrade overall performance, whereas we aim to change the prediction of a specific unmodified target. Blum et al. (2021) theoretically characterizes when correctly labeled injected examples can induce errors on particular test points as a function of the hypothesis class, data distribution, and poisoning budget, while Lu et al. (2023) studies the reachability of adversarial model behavior under constrained poisoning. Hidden Poison Di et al. (2023) considers targeted clean-label poisoning in a machine-unlearning setting, where camouflage removal activates the attack after training. Our setting instead assumes standard victim training on a single poisoned dataset without subsequent intervention. Recent advancements in targeted clean-label attacks have also explored first-order efficient approximations for bilevel optimization Zheng & Li (2021) and mechanisms that induce dominant features Zhang et al. (2023). Our work complements these attack formulations by focusing on target-conditioned base selection prior to optimization.

The choice of which examples to poison has been analyzed for clean-label backdoors. FUS Xia et al. (2022) uses forgetting behavior to identify effective poisons, while Gao et al. (2023) selects hard target-class examples based on loss, gradient norm, and forgetting statistics. Sleeper Agent Souri et al. (2022) likewise exploits variation among candidate training examples when constructing hidden triggers. Hung-Quang et al. (2023) replaces random poisoning with selective poisoning when only limited target-class information is available. Wicked Oddities Nguyen et al. (2025) extends this by detecting out-of-distribution samples via pretrained representations. Wu et al. (2026) further unifies sample selection and trigger construction, making selection depend on trigger and trigger-implanted sample properties, while Zhang et al. (2025) uses sample-specific feature alignment for clean-label backdoor construction. Related evidence also appears in text classification, where Kirci & Gursoy (2025) improves clean-label backdoor attacks by poisoning misclassified examples. These works show that poison-set composition affects attack effectiveness, but their criteria aim to reinforce a reusable trigger or target-class behavior. In contrast, our selector is conditioned on the relationship between a candidate clean base and one specific unmodified test target. A sample may be difficult, influential, or helpful for learning a trigger while providing little useful training signal for that specific target.

This distinction is also related to earlier work on example influence and poison importance. Influence functions Koh & Liang (2017) quantify how individual training examples affect particular predictions and can be used to construct targeted training perturbations. We instead use target-conditioned quantities before perturbation optimization to determine which clean examples should be passed to a downstream poison optimizer. Yang et al. (2022) exploit variation in poison influence defensively by suppressing particularly harmful examples, while Friendly Noise Liu et al. (2022) studies how poison signals interact with victim optimization and introduces training-time perturbations to reduce their effect. Poison Forensics Shan et al. (2022) addresses the retrospective

problem of identifying training examples responsible for an observed poisoned prediction. Other defensive work explores feature-map anomaly detection to filter out clean-label poisons De Gaspari et al. (2024). Guided Diffusion Poisoning (GDP) Souri et al. (2024) is closest to our focus on initialization: it synthesizes favorable poison bases with a diffusion model before downstream poison optimization. Our setting instead selects favorable bases from the existing clean training set, without generating new examples.

Overall, prior work shows that poisoning effectiveness depends on which training examples are modified, their geometry and influence, and, in backdoor settings, their interaction with the trigger. Our formulation isolates the complementary problem of selecting existing adversarial-class examples for a particular target before any poison perturbations are optimized. The selected bases are then passed to the downstream attack without changing its objective, allowing base quality and poison construction to be studied separately.

C GRADIENT ALIGNMENT FOR BASE-SAMPLE SELECTION

We provide a detailed derivation of the gradient-alignment expression used in Section 3. Consider a clean candidate sample x_i with label y_{adv} and the target sample x_t . At a fixed model state θ , define their parameter gradients as

$$g_i = \nabla_{\theta} \ell(x_i, y_{\text{adv}}; \theta), \quad g_t = \nabla_{\theta} \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta).$$

Both losses are evaluated with respect to the adversarial class y_{adv} , since the attack seeks to make x_t be predicted as this class rather than its ground-truth class y_t . We first show how the candidate gradient affects the adversarial loss at the target sample. Consider a sufficiently small gradient-descent update induced by x_i , given by $\theta' = \theta - \eta g_i$, where $\eta > 0$. A first-order Taylor expansion of the adversarial loss around θ gives

$$\mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta') = \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta) + \langle \nabla_{\theta} \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta), \theta' - \theta \rangle + \mathcal{O}(\|\theta' - \theta\|_2^2).$$

Substituting $\theta' - \theta = -\eta g_i$ and using the definition of g_t , the first-order term becomes

$$\langle \nabla_{\theta} \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta), \theta' - \theta \rangle = \langle g_t, -\eta g_i \rangle = -\eta \langle g_t, g_i \rangle.$$

Moreover, $\|\theta' - \theta\|_2^2 = \eta^2 \|g_i\|_2^2$, and therefore the remainder is $\mathcal{O}(\eta^2)$. Hence,

$$\mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta') = \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta) - \eta \langle g_i, g_t \rangle + \mathcal{O}(\eta^2).$$

Since $\eta > 0$, a positive $\langle g_i, g_t \rangle$ produces a negative first-order change in the adversarial loss. A larger positive inner product therefore indicates that the update induced by x_i is more strongly aligned with decreasing the attack objective at x_t .

We next expose the structure of this inner product. We separate the representation network from the final linear classifier and write $z_{\theta}(x) = Wh_{\phi}(x)$ and $\theta = (W, \phi)$, where $h_{\phi}(x) \in \mathbb{R}^d$ is the representation before the classifier and $W \in \mathbb{R}^{C \times d}$ contains the classifier weights. A bias term, if present, can be absorbed into this expression by appending a constant coordinate to $h_{\phi}(x)$. We use $h_i = h_{\phi}(x_i)$ and $h_t = h_{\phi}(x_t)$.

Define the gradients of the candidate and adversarial losses with respect to their logits as

$$r_i = \nabla_{z_{\theta}(x_i)} \ell(x_i, y_{\text{adv}}; \theta), \quad r_t = \nabla_{z_{\theta}(x_t)} \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta).$$

For softmax cross-entropy, let $p_{\theta}(x) = \text{softmax}(z_{\theta}(x))$. For an arbitrary input x evaluated against y_{adv} , the loss can be written as

$$\ell = -\log p_{\theta, y_{\text{adv}}}(x) = -z_{\theta, y_{\text{adv}}}(x) + \log \left(\sum_{c=1}^C \exp(z_{\theta, c}(x)) \right).$$

Differentiating with respect to a logit $z_{\theta, c}(x)$ gives

$$\frac{\partial \ell}{\partial z_{\theta, c}(x)} = -\mathbf{1}[c = y_{\text{adv}}] + \frac{\exp(z_{\theta, c}(x))}{\sum_{k=1}^C \exp(z_{\theta, k}(x))} = p_{\theta, c}(x) - \mathbf{1}[c = y_{\text{adv}}].$$

Collecting the derivatives over all C logits yields $\nabla_z \ell = p_\theta(x) - e_{\text{adv}}$, where e_{adv} is the one-hot vector corresponding to y_{adv} . Thus, under softmax cross-entropy, $r_i = p_\theta(x_i) - e_{\text{adv}}$ and $r_t = p_\theta(x_t) - e_{\text{adv}}$. We now propagate these logit gradients through the final classifier. Let $w_c^\top$ denote row c of W . Since $z_{\theta,c}(x) = w_c^\top h_\phi(x) = \sum_{j=1}^d w_{c,j} h_j$, we have $\partial z_{\theta,c} / \partial w_{c,j} = h_j$, while $\partial z_{\theta,c'} / \partial w_{c,j} = 0$ for $c' \neq c$. Therefore,

$$\frac{\partial \ell}{\partial w_{c,j}} = \sum_{c'=1}^C \frac{\partial \ell}{\partial z_{\theta,c'}} \frac{\partial z_{\theta,c'}}{\partial w_{c,j}} = \frac{\partial \ell}{\partial z_{\theta,c}} h_j = r_c h_j.$$

Writing $r = \nabla_z \ell$ and $h = h_\phi(x)$, collecting all coordinates gives the classifier-weight gradient

$$\nabla_W \ell = r h^\top.$$

Applying this result to the candidate and target gives

$$\nabla_W \ell(x_i, y_{\text{adv}}; \theta) = r_i h_i^\top, \quad \nabla_W \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta) = r_t h_t^\top.$$

We next obtain the gradients with respect to the representation parameters ϕ . Since $z = Wh$, for each representation coordinate h_j ,

$$\frac{\partial \ell}{\partial h_j} = \sum_{c=1}^C \frac{\partial \ell}{\partial z_c} \frac{\partial z_c}{\partial h_j} = \sum_{c=1}^C r_c w_{c,j},$$

which in vector form gives $\nabla_h \ell = W^\top r$. Because $h_\phi(x)$ depends on ϕ , applying the chain rule once more gives

$$\nabla_\phi \ell = \left(\frac{\partial h_\phi(x)}{\partial \phi} \right)^\top \nabla_h \ell = \left(\frac{\partial h_\phi(x)}{\partial \phi} \right)^\top W^\top r.$$

Therefore, for the candidate and target,

$$\nabla_\phi \ell(x_i, y_{\text{adv}}; \theta) = \left(\frac{\partial h_\phi(x_i)}{\partial \phi} \right)^\top W^\top r_i, \quad \nabla_\phi \mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta) = \left(\frac{\partial h_\phi(x_t)}{\partial \phi} \right)^\top W^\top r_t.$$

Since $\theta = (W, \phi)$, the full parameter gradient consists of the classifier-weight and representation-parameter components. Consequently, their inner product can be separated as

$$\langle g_i, g_t \rangle = \langle \text{vec}(r_i h_i^\top), \text{vec}(r_t h_t^\top) \rangle + \left\langle \left(\frac{\partial h_\phi(x_i)}{\partial \phi} \right)^\top W^\top r_i, \left(\frac{\partial h_\phi(x_t)}{\partial \phi} \right)^\top W^\top r_t \right\rangle.$$

The first term is the Frobenius inner product between the two classifier gradients:

$$\langle r_i h_i^\top, r_t h_t^\top \rangle_F = \text{tr} [(r_i h_i^\top)^\top (r_t h_t^\top)] = \text{tr} [h_i r_i^\top r_t h_t^\top] = (r_i^\top r_t) (h_i^\top h_t).$$

For the representation-parameter term

$$\begin{aligned} \left\langle \left(\frac{\partial h_\phi(x_i)}{\partial \phi} \right)^\top W^\top r_i, \left(\frac{\partial h_\phi(x_t)}{\partial \phi} \right)^\top W^\top r_t \right\rangle &= \left[\left(\frac{\partial h_\phi(x_i)}{\partial \phi} \right)^\top W^\top r_i \right]^\top \left[\left(\frac{\partial h_\phi(x_t)}{\partial \phi} \right)^\top W^\top r_t \right] \\ &= r_i^\top W \frac{\partial h_\phi(x_i)}{\partial \phi} \left(\frac{\partial h_\phi(x_t)}{\partial \phi} \right)^\top W^\top r_t. \end{aligned}$$

Combining the two contributions gives

$$\begin{aligned} \langle g_i, g_t \rangle &= (r_i^\top r_t) (h_i^\top h_t) + r_i^\top W \frac{\partial h_\phi(x_i)}{\partial \phi} \left(\frac{\partial h_\phi(x_t)}{\partial \phi} \right)^\top W^\top r_t \\ &= r_i^\top \left[(h_i^\top h_t) I + W \frac{\partial h_\phi(x_i)}{\partial \phi} \left(\frac{\partial h_\phi(x_t)}{\partial \phi} \right)^\top W^\top \right] r_t, \end{aligned}$$

where $I \in \mathbb{R}^{C \times C}$ is the identity matrix. This is Eq. (3) in the main text.

D IMPLEMENTATION DETAILS

This appendix outlines the experimental setup for the main evaluation in Section 4.1. Unless noted otherwise, RAND and GRAFT use the same data, target images, model initialization, poison budgets, attack hyperparameters, and victim training procedures. They differ only in the clean training examples chosen as poison bases.

Datasets. All main experiments use CIFAR-10 with its standard training and test set. Images are normalized channel-wise using $\mu = (0.4914, 0.4822, 0.4465)$ and $\sigma = (0.2023, 0.1994, 0.2010)$. Base selection and model training operate on normalized inputs. Before poison optimization, the selected bases are mapped back to the $[0, 1]$ pixel domain. Perturbations are optimized in this domain, with the resulting images clipped to $[0, 1]$ after each update. The final poisons are normalized again before being inserted into the victim training set. We use the poison-target convention for class-pair notation. Therefore, dog→bird denotes dog as the poison class y_{adv} and bird as the target class. Poisoned examples maintain their original label y_{adv} throughout victim training.

Model Training. We evaluate ConvNet, ResNet20, and VGG13. ConvNet consists of three convolutional stages with a width of 128, ReLU activations, batch normalization, and average pooling. ResNet20 and VGG13 also use their batch-normalized implementations. All surrogate and victim models are trained from scratch using cross-entropy loss and SGD with a momentum of 0.9. Each surrogate used in experiments is trained for 60 epochs with a batch size of 128, and learning-rate reductions at epochs 35 and 45. No data augmentation is used during surrogate training. For each target, we train six independently initialized victim models from scratch on the poisoned training set. Victims are trained for 50 epochs with a learning rate of 0.1, a batch size of 125, and a momentum of 0.9. No victim-side data augmentation is used in the main evaluation.

Target Selection. Targets are selected from CIFAR-10 test examples belonging to the target class. We exclude examples that the clean ranking ensemble already predicts as y_{adv} , since these would constitute successful targeted predictions before poisoning. For each model, attack, and class pair, we evaluate ten targets. Once selected, the target indices are fixed and reused across all base-selection variants and poison budgets. RAND and GRAFT therefore attack exactly the same target images. The target indices used in the main evaluation are reported in Table 4.

Table 4: Target images used in the main evaluation. The same ten CIFAR-10 test images are used across base-selection variants and poison budgets for each model, class pair, and attack.

Model	Pair	Attack	CIFAR-10 test indices
ConvNet	frog→airplane	BP	2540, 8129, 3725, 5705, 7663, 3875, 5169, 6335, 7488, 9831
ConvNet	frog→airplane	GM	843, 1257, 8252, 8730, 8055, 5859, 8954, 821, 1547, 4427
ConvNet	frog→airplane	BP	7511, 857, 7010, 2091, 1776, 1596, 5354, 3158, 6006, 6191
ConvNet	frog→airplane	GM	496, 2087, 8941, 9820, 3825, 1796, 2548, 6437, 3336, 2603
ResNet20	frog→airplane	BP	3495, 2903, 6697, 2315, 6899, 2312, 6353, 4421, 2242, 6768
ResNet20	frog→airplane	GM	5118, 8740, 5324, 4630, 833, 2065, 7056, 9059, 6707, 9874
ResNet20	frog→airplane	BP	9273, 2535, 4118, 5462, 8476, 3342, 313, 4490, 2672, 6481
ResNet20	frog→airplane	GM	9235, 338, 255, 3207, 8563, 912, 4639, 2570, 3117, 2041
VGG13	frog→airplane	BP	3728, 5640, 6922, 7855, 2127, 7586, 7360, 2171, 4462, 2597
VGG13	frog→airplane	GM	630, 409, 2270, 9870, 1656, 5022, 9090, 6033, 7940, 9731
VGG13	frog→airplane	BP	912, 9957, 1739, 1165, 3278, 4313, 416, 6491, 338, 4287
VGG13	frog→airplane	GM	2783, 2507, 6738, 5787, 7815, 5074, 7356, 8267, 2264, 6481

Perturbation Optimization. We provide the main optimization details for the three poisoning objectives used in our evaluation

- **Bullseye Polytope (BP)** (Aghakhani et al., 2021): We minimize the distance between the target representation and the mean representation of the poison set across the crafting surrogates. BP is optimized for 250 steps using sign-gradient updates with a step size of $1/255$. Perturbations are initialized uniformly within the feasible region and projected onto the $8/255$ constraint after each update.
- **Gradient Matching (GM)** (Geiping et al., 2020): We match the gradient induced by the poison set to the gradient of the target evaluated with the adversarial label. For crafting surrogate

k , let $g_t^{(k)}$ and $g_p^{(k)}$ denote the target and poison gradients, respectively. We minimize the average cosine mismatch:

$$\mathcal{L}_{\text{GM}} = \frac{1}{K_c} \sum_{k=1}^{K_c} \left(1 - \frac{\langle g_p^{(k)}, g_t^{(k)} \rangle}{\|g_p^{(k)}\|_2 \|g_t^{(k)}\|_2}\right), \quad K_c = 5.$$

Perturbations are optimized with Adam using a learning rate of $1/255$ and are projected back onto the feasible region after every update.

- **Sharpness-Aware Poisoning (SAPA)** (He et al., 2024): This follows the same poison optimization procedure as GM but replaces the standard target gradient with a sharpness-aware target gradient. For each crafting surrogate, we compute the target gradient $g_t^{(k)}$ and evaluate the matching objective at the perturbed model parameters:

$$\theta' = \theta + \gamma \frac{g_t^{(k)}}{\|g_t^{(k)}\|_2}, \quad \gamma = 0.05.$$

E ADDITIONAL RESULTS AND STATISTICAL ANALYSIS

The main evaluation in Table 1 considers ten attack targets, with six independently initialized victim models trained for each target. While the main table reports ASR, Figure 1 shows CTA only for a subset of the evaluated settings. We therefore report the complete CTA results in Table 5 and complement the main ASR results with a target-level statistical analysis in Table 6. The complete CTA results show that the improvement in attack success does not come with a comparable reduction in clean performance. Across the 108 configurations, the mean CTA decreases by only 0.06% for GRAFT and 0.03% for GRAFT⁺ relative to RAND. The largest absolute difference is below 0.7%. A modest decrease becomes more visible at the largest poisoning budget, $\rho = 0.04$, where CTA is lower than RAND by 0.27 percentage on average for GRAFT and GRAFT⁺, respectively. Overall, clean accuracy remains largely unchanged despite substantially different attack success rates.

For the statistical analysis, the six victim runs corresponding to a given target share the same target image and optimized poison set. They therefore should not be treated as six independent observations. Instead, we use the attack target as the unit of analysis. For method v , the success rate of target t is first computed over its six victim-training runs, and the reported ASR is then obtained by averaging these rates across the ten targets,

$$r_t^{(v)} = \frac{1}{6} \sum_{s=1}^6 a_{t,s}^{(v)} \quad \text{ASR}^{(v)} = \frac{1}{10} \sum_{t=1}^{10} r_t^{(v)},$$

where $a_{t,s}^{(v)}$ is one when the attack succeeds and zero otherwise. The standard error reported in parentheses in Table 6 is computed across these ten target-level rates and therefore reflects variation in attack success from one target to another. Because RAND and GRAFT⁺ are evaluated on the same targets, we compare them in a paired manner. For each target, we compute the difference between their success rates and average these differences as

$$d_t = r_t^{(\text{GRAFT}^+)} - r_t^{(\text{RAND})}, \quad \Delta = \frac{1}{10} \sum_{t=1}^{10} d_t.$$

Therefore, Δ measures the average ASR improvement of GRAFT⁺ over RAND in percentage. The 95% confidence interval in Table 6 describes the uncertainty around this improvement rather than around either ASR individually. Intuitively, it indicates how much the measured GRAFT⁺-RAND difference could vary if the evaluation were repeated with a different sample of attack targets. An interval entirely above zero indicates positive improvement for GRAFT⁺; an interval including zero means the advantage varies enough across targets that no improvement is also consistent with the sample. An interval entirely below zero favors RAND. We estimate these intervals using a paired bootstrap over targets. Each bootstrap replicate samples ten targets with replacement from the original ten. Whenever a target is selected, all six victim outcomes for that target are kept together for both selection methods, preserving both the dependence among victim runs and the pairing between

RAND and GRAFT⁺. We recompute Δ for each resampled target set and use the 2.5th and 97.5th percentiles of the resulting distribution as the 95% confidence interval.

Table 6 shows that the average ASR increases from 31.2% for RAND to 57.7% for GRAFT⁺. Among the 108 configurations, 70 confidence intervals lie entirely above zero, 36 include zero, and only two lie entirely below zero. The evidence is particularly consistent for GM and SAPA, for which 29 of 36 and 28 of 36 intervals are entirely positive, respectively. The two intervals entirely below zero occur for ResNet20 with BP at the largest poisoning budget, consistent with RAND increasingly matching GRAFT at larger poisoning budgets. Overall, the target-level analysis shows that the improvement is broad across configurations while also making clear where its magnitude is more dependent on the particular attack targets.

Table 5: CTA results corresponding to the ASR evaluation in Table 1. Entries report CTA (%), $\uparrow$.

ρ ($\times 10^{-3}$)	dog $\rightarrow$ bird									frog $\rightarrow$ airplane								
	BP			GM			SAPA			BP			GM			SAPA		
	RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺	RAND	GRAFT	GRAFT ⁺
ConvNet	1	80.14	80.18	80.14	80.34	80.29	80.56	80.43	80.32	80.50	80.04	80.09	80.06	80.24	80.24	80.29	80.18	80.19
	2	80.13	80.12	80.13	80.38	80.38	80.39	80.39	80.42	80.42	80.10	80.04	80.04	80.16	80.17	80.16	80.15	80.18
	5	80.11	80.22	80.22	80.46	80.41	80.47	80.30	80.40	80.47	80.06	80.04	79.99	80.20	80.03	80.22	80.16	80.18
	10	80.10	80.14	80.23	80.46	80.57	80.59	80.36	80.43	80.55	80.09	80.02	80.11	80.17	80.17	80.14	80.14	80.10
	40	80.09	80.11	80.13	80.38	80.40	80.53	80.31	80.39	80.57	80.02	79.91	79.94	80.10	80.05	80.15	80.10	80.05
ResNet20	1	83.80	83.81	83.74	83.76	83.61	83.86	83.76	83.71	83.78	83.75	83.67	83.88	83.80	83.65	83.78	83.75	83.68
	2	83.73	83.68	83.73	83.88	83.79	83.74	83.63	83.78	83.72	83.76	83.74	83.68	83.77	83.79	83.80	83.79	83.77
	5	83.71	83.73	83.77	83.79	83.81	83.80	83.85	83.89	83.71	83.69	83.72	83.71	83.75	83.73	83.81	83.80	83.79
	10	83.62	83.62	83.76	83.73	83.70	83.81	83.74	83.76	83.75	83.86	83.72	83.69	83.72	83.69	83.75	83.68	83.78
	40	83.56	83.72	83.81	83.67	83.72	83.81	83.62	83.64	83.84	83.72	83.66	83.55	83.57	83.45	83.58	83.67	83.45
VGG13	1	85.34	85.45	85.35	85.52	85.33	85.43	85.53	85.50	85.36	85.34	85.36	85.32	85.21	85.23	85.15	85.43	85.19
	2	85.25	85.30	85.44	85.54	85.46	85.44	85.59	85.55	85.42	85.34	85.47	85.28	85.23	85.37	85.36	85.26	85.25
	5	85.42	85.37	85.56	85.49	85.46	85.50	85.49	85.47	85.46	85.30	85.39	85.31	85.20	85.30	85.41	85.39	85.16
	10	85.30	85.30	85.27	85.45	85.52	85.42	85.36	85.29	85.46	85.36	85.26	85.35	85.35	85.19	85.17	85.21	85.12
	40	85.32	85.29	85.22	85.29	85.25	85.43	85.33	85.30	85.25	85.45	85.30	85.03	85.11	85.09	85.14	85.23	85.00
	40	85.02	84.92	84.71	85.05	84.86	84.93	85.04	84.83	84.84	85.24	84.71	84.56	84.99	84.62	84.69	85.00	84.59

F BASE SELECTION ACROSS ADDITIONAL CLASS PAIRS

The main evaluation in Section 4.1 focuses on two target-adversarial class pairs, so we next test whether the benefit of informed base selection extends to a broader range of attack instances. Table 7 reports results on six additional CIFAR-10 class pairs using ConvNet at $\epsilon = 5 \times 10^{-3}$. For each pair, we evaluate BP, GM, and SAPA using five fixed targets and five independently trained victims per target. The class pairs are fixed before attack evaluation. We also report bird $\rightarrow$ dog under the same protocol as a reference to one of the class pairs considered in the main evaluation.

Across the six additional pairs, DPP improves ASR from 12.7% to 64.7% for BP, from 12.0% to 46.7% for GM, and from 14.0% to 44.7% for SAPA. DPP achieves higher observed ASR than Random in 15 of the 18 pairs and attack combinations and ties in the remaining three. Several gains are substantial even when random base selection produces a weak attack. For example, deer $\rightarrow$ horse increases from 8.0% to 100.0% with BP and from 4.0% to 72.0% with both GM and SAPA. Similarly, bird $\rightarrow$ airplane improves from 24.0% to 88.0% with BP and from 12.0% to 80.0% with GM. At the same time, the results retain clear variation across attack instances. In particular, ship $\rightarrow$ frog remains difficult for all three objectives, while truck $\rightarrow$ cat remains ineffective for GM despite informed selection. Base selection therefore does not eliminate differences in the intrinsic difficulty of individual target and adversarial class combinations. Instead, the consistent improvements across most of the additional pairs show that the gains observed in the main evaluation are not restricted to its original class pairs and that the choice of attack initialization remains important across substantially different targeted poisoning instances.

Table 6: Target-level statistical analysis of the main ASR results for ConvNet, ResNet20, and VGG13 in Table 1. RAND and GRAFT⁺ entries report mean ASR (%) across ten attack targets, with the standard error across target-level success rates in parentheses. The 95% CI columns report paired bootstrap confidence intervals for the GRAFT⁺-RAND ASR difference in percentage.

	ρ ($\times 10^{-3}$)	BP			GM			SAPA		
		RAND ASR (SE)	GRAFT ⁺ ASR (SE)	95% CI GRAFT ⁺ -RAND	RAND ASR (SE)	GRAFT ⁺ ASR (SE)	95% CI GRAFT ⁺ -RAND	RAND ASR (SE)	GRAFT ⁺ ASR (SE)	95% CI GRAFT ⁺ -RAND
ConvNet										
dog→bird	1	0.0 (0.0)	41.7 (8.3)	[26.7, 56.7]	0.0 (0.0)	45.0 (9.1)	[27.5, 60.0]	0.0 (0.0)	31.2 (10.2)	[14.6, 50.0]
	2	1.7 (1.7)	70.0 (7.4)	[55.0, 81.7]	0.0 (0.0)	53.3 (6.7)	[38.8, 61.7]	0.0 (0.0)	45.8 (7.8)	[31.2, 59.6]
	5	13.3 (6.5)	95.0 (2.5)	[68.3, 91.7]	1.7 (1.7)	61.7 (10.3)	[41.2, 77.5]	6.7 (3.7)	71.2 (11.3)	[42.1, 78.8]
	10	35.0 (8.0)	95.0 (2.5)	[43.3, 71.7]	21.7 (7.9)	87.9 (7.5)	[41.3, 83.3]	23.3 (7.5)	88.8 (5.9)	[42.1, 76.2]
	20	73.3 (9.7)	93.3 (3.7)	[6.7, 35.0]	43.3 (9.4)	100.0 (0.0)	[10.4, 45.8]	50.0 (10.5)	100.0 (0.0)	[22.9, 62.5]
	40	91.7 (3.7)	100.0 (0.0)	[1.7, 15.0]	83.3 (10.5)	100.0 (0.0)	[0.0, 6.2]	76.7 (10.3)	100.0 (0.0)	[2.1, 25.0]
frog→airplane	1	0.0 (0.0)	3.3 (2.2)	[0.0, 8.3]	0.0 (0.0)	5.0 (3.3)	[0.0, 12.5]	0.0 (0.0)	2.5 (2.5)	[0.0, 7.5]
	2	0.0 (0.0)	10.0 (5.1)	[0.0, 20.0]	0.0 (0.0)	5.0 (3.6)	[0.0, 13.3]	0.0 (0.0)	10.0 (5.7)	[0.0, 21.7]
	5	0.0 (0.0)	26.7 (8.7)	[11.7, 43.3]	0.0 (0.0)	15.0 (6.3)	[3.3, 26.7]	0.0 (0.0)	11.7 (5.0)	[3.3, 21.7]
	10	1.7 (1.7)	45.0 (12.7)	[18.3, 61.7]	3.3 (2.2)	16.7 (9.9)	[0.0, 31.7]	1.7 (1.7)	6.7 (5.1)	[0.0, 13.3]
	20	10.0 (4.4)	55.0 (12.7)	[23.3, 66.7]	3.3 (3.3)	47.5 (10.0)	[26.7, 58.3]	8.3 (5.1)	35.0 (12.4)	[9.2, 40.4]
	40	30.0 (10.2)	63.3 (13.1)	[20.0, 46.7]	16.7 (7.9)	55.0 (13.0)	[25.8, 68.8]	25.0 (8.3)	67.5 (13.6)	[20.4, 64.6]
ResNet20										
dog→bird	1	11.7 (5.0)	12.5 (3.7)	[-12.5, 7.1]	15.0 (5.2)	40.0 (13.1)	[7.1, 47.1]	18.3 (6.3)	37.5 (9.6)	[10.8, 40.0]
	2	16.7 (5.0)	37.5 (10.3)	[2.1, 46.2]	31.7 (8.4)	60.0 (14.1)	[15.0, 58.3]	21.7 (7.9)	67.5 (10.0)	[40.0, 65.4]
	5	30.0 (7.8)	27.5 (8.4)	[-15.8, 16.7]	43.3 (7.5)	82.5 (8.0)	[21.7, 59.2]	55.0 (10.6)	75.0 (7.3)	[11.2, 46.7]
	10	43.3 (10.3)	50.0 (11.7)	[-15.0, 26.7]	56.7 (11.4)	85.0 (10.5)	[15.8, 55.0]	65.0 (8.8)	85.0 (6.3)	[17.9, 38.7]
	20	55.0 (8.3)	33.3 (14.8)	[-33.3, 2.1]	75.0 (8.7)	100.0 (0.0)	[14.6, 50.0]	95.0 (3.6)	100.0 (0.0)	[0.0, 16.7]
	40	48.3 (12.3)	18.8 (11.1)	[-29.2, -8.3]	98.3 (1.7)	100.0 (0.0)	[0.0, 0.0]	98.3 (1.7)	100.0 (0.0)	[0.0, 10.0]
frog→airplane	1	8.3 (3.7)	12.5 (5.3)	[-3.3, 12.1]	1.7 (1.7)	22.5 (8.0)	[7.5, 35.4]	1.7 (1.7)	32.5 (7.5)	[17.5, 45.0]
	2	8.3 (3.7)	5.0 (3.3)	[-10.4, 5.4]	0.0 (0.0)	26.7 (8.3)	[11.7, 41.7]	5.0 (3.6)	35.0 (8.8)	[11.7, 48.3]
	5	18.3 (5.8)	25.0 (5.0)	[-2.9, 15.0]	8.3 (3.7)	61.7 (7.5)	[40.0, 66.7]	3.3 (2.2)	63.3 (10.5)	[40.0, 80.0]
	10	16.7 (5.0)	22.5 (8.0)	[-6.7, 24.2]	26.7 (9.7)	68.5 (9.4)	[27.8, 61.1]	21.7 (6.6)	75.0 (8.5)	[49.2, 67.9]
	20	30.0 (10.2)	41.7 (11.4)	[-6.3, 27.1]	45.0 (8.6)	85.4 (5.8)	[22.9, 47.9]	40.0 (7.1)	85.7 (7.7)	[38.1, 61.9]
	40	33.3 (10.8)	25.0 (10.9)	[-20.8, -6.2]	70.0 (7.8)	97.6 (2.4)	[9.5, 50.0]	81.7 (5.2)	94.4 (3.5)	[8.3, 30.6]
VGG13										
dog→bird	1	21.7 (6.1)	30.0 (7.6)	[-5.8, 20.8]	13.3 (6.0)	70.0 (6.5)	[35.0, 70.0]	11.7 (5.6)	87.5 (5.3)	[59.2, 88.8]
	2	41.7 (10.3)	37.5 (9.6)	[-32.5, 21.3]	21.7 (9.3)	77.5 (7.0)	[34.6, 70.4]	26.7 (9.0)	87.5 (6.5)	[36.3, 78.3]
	5	45.0 (7.9)	55.0 (9.1)	[-10.0, 15.8]	55.0 (9.3)	100.0 (0.0)	[22.9, 64.6]	60.0 (9.7)	95.0 (3.3)	[8.8, 53.3]
	10	41.7 (8.7)	55.0 (8.6)	[0.0, 25.0]	76.7 (10.0)	100.0 (0.0)	[2.1, 41.7]	88.3 (5.6)	100.0 (0.0)	[2.1, 16.7]
	20	61.7 (11.7)	56.2 (10.9)	[-25.0, 12.5]	96.7 (3.3)	100.0 (0.0)	[0.0, 12.5]	100.0 (0.0)	100.0 (0.0)	[0.0, 0.0]
	40	75.0 (5.7)	60.4 (11.8)	[-31.3, 0.0]	98.3 (1.7)	100.0 (0.0)	[0.0, 10.0]	100.0 (0.0)	100.0 (0.0)	[0.0, 0.0]
frog→airplane	1	1.7 (1.7)	7.5 (5.3)	[-4.2, 17.5]	0.0 (0.0)	40.0 (6.5)	[27.5, 50.0]	0.0 (0.0)	45.0 (13.5)	[20.0, 70.0]
	2	1.7 (1.7)	12.5 (6.5)	[2.5, 20.8]	5.0 (3.6)	56.7 (12.0)	[28.3, 73.3]	10.0 (4.4)	51.7 (8.8)	[28.3, 53.3]
	5	3.3 (2.2)	12.5 (7.5)	[-3.3, 27.5]	16.7 (6.6)	80.0 (9.6)	[43.3, 81.7]	15.0 (6.8)	70.0 (10.5)	[36.7, 73.3]
	10	13.3 (4.8)	20.0 (10.0)	[-3.3, 33.8]	28.3 (8.6)	80.0 (7.8)	[33.3, 70.0]	20.0 (8.5)	80.8 (9.1)	[39.2, 80.8]
	20	10.0 (6.7)	31.2 (12.4)	[8.3, 50.0]	56.7 (10.3)	85.4 (12.4)	[10.4, 43.8]	53.3 (9.9)	91.7 (8.3)	[20.8, 54.2]
	40	16.7 (8.6)	10.4 (6.2)	[-16.7, 8.3]	78.3 (10.3)	87.5 (12.5)	[2.1, 16.7]	85.0 (10.1)	100.0 (0.0)	[0.0, 36.7]

F.1 EVALUATION ON ADDITIONAL DATASETS

We first test whether the benefit of diversity-aware base selection extends beyond CIFAR-10. CIFAR-100 provides a more constrained setting with many more classes and only 500 training candidates per adversarial class, and therefore uses ResNet18BN. SVHN provides a complementary shift from natural images to street-view digits while retaining the 32×32 resolution and the ConvNetBN backbone. Table 8 reports the original five-instance comparison. DPP improves ASR for both GM and SAPA on all three datasets. Absolute ASR values should not be compared across dataset rows because the number of classes, candidate-pool size, and backbone differ.

The preceding comparison holds the attack and budget fixed, so we additionally evaluate a broader, deliberately non-exhaustive grid in Table 9. This grid compares uniform Random selection with the plain Greedy selector; Greedy uses 20 selector surrogates, cosine distance, and no Jacobian term. It separates two sources of variation: a budget slice on one fixed pair and a class-pair slice at the middle budget. All budgets are below 0.01999. In CIFAR-100, the largest budget is 0.005 (250 of the 500 adversarial-class candidates), avoiding the degenerate regime in which selection exhausts

Table 7: Attack success across additional CIFAR-10 target–adversarial class pairs. Experiments use ConvNet with $\epsilon = 5 \times 10^{-3}$. Entries report ASR (%) averaged across all targets and victim runs. The final row reports bird→dog under the same protocol for reference.

Target → Adv.	BP		GM		SAPA	
	Random	DPP	Random	DPP	Random	DPP
cat → dog	40.0	100.0	44.0	80.0	36.0	80.0
deer → horse	8.0	100.0	4.0	72.0	4.0	72.0
automobile → truck	4.0	68.0	12.0	48.0	12.0	48.0
bird → airplane	24.0	88.0	12.0	80.0	32.0	64.0
ship → frog	0.0	4.0	0.0	0.0	0.0	0.0
truck → cat	0.0	28.0	0.0	0.0	0.0	4.0
Average	12.7	64.7	12.0	46.7	14.0	44.7
bird → dog	24.0	80.0	12.0	64.0	24.0	68.0

Table 8: Evaluation of Random and DPP base selection on CIFAR-10, CIFAR-100, and SVHN. Results report ASR over five fixed attack instances with five victim runs per target.

Dataset (Model)	GM		SAPA	
	Random	DPP	Random	DPP
CIFAR-10 (ConvNetBN)	12.0	64.0	24.0	68.0
CIFAR-100 (ResNet18BN)	12.0	24.0	4.0	16.0
SVHN (ConvNetBN)	36.0	80.0	28.0	88.0

the candidate pool. For Tiny ImageNet we use ResNet18BN at 64×64 rather than ConvNetBN and restrict the slice to 0.0005–0.002 for the same candidate-pool reason.

Each cell below is an independent experiment. CIFAR-100 and Tiny ImageNet use four pinned targets with four victim seeds per target, while SVHN uses six pinned targets with five victim seeds per target. Target sets are shared by Random and Greedy, by FC, GM, and SAPA, and by every budget for a given class pair. This makes the selection methods directly paired within every attack. Entries report ASR (%) with clean test accuracy in parentheses; “–” denotes a pending cell.

Table 9: Broader additional-dataset ablation comparing Random and plain Greedy base selection, without a Jacobian score. The pair is written target $\rightarrow$ adversarial class and the budget column is in units of 10^{-3} . CIFAR-100 and Tiny ImageNet cells use four targets and four victims per target; SVHN cells use six targets and five victims per target.

Dataset	Model	Target $\rightarrow$ adversarial	Budget	FC		GM		SAPA	
				Random	Greedy	Random	Greedy	Random	Greedy
CIFAR-100	ResNet18BN	willow tree $\rightarrow$ sea	1	6.3 (57.90)	6.3 (57.83)	12.5 (57.16)	37.5 (57.82)	12.5 (57.94)	25.0 (57.77)
		willow tree $\rightarrow$ sea	2	0.0 (57.84)	6.3 (57.58)	43.8 (57.98)	43.8 (57.03)	50.0 (57.66)	50.0 (57.53)
		willow tree $\rightarrow$ sea	5	0.0 (57.89)	12.5 (57.98)	93.8 (57.47)	100.0 (58.26)	81.3 (57.37)	75.0 (57.36)
		bicycle $\rightarrow$ plain	2	0.0 (58.04)	0.0 (56.98)	0.0 (58.09)	12.5 (57.74)	0.0 (57.56)	0.0 (57.88)
		lawn mower $\rightarrow$ wardrobe	2	6.3 (58.16)	6.3 (57.46)	6.3 (58.13)	25.0 (58.05)	0.0 (58.03)	18.8 (57.29)
		road $\rightarrow$ bottle	2	0.0 (57.80)	0.0 (57.97)	0.0 (57.57)	31.2 (57.39)	0.0 (57.16)	12.5 (57.58)
		cattle $\rightarrow$ sunflower	2	0.0 (57.52)	6.3 (58.05)	6.3 (57.16)	25.0 (57.91)	0.0 (57.74)	12.5 (58.38)
SVHN	ConvNetBN	2 $\rightarrow$ 9	1	3.3 (92.18)	33.3 (92.19)	0.0 (92.20)	63.3 (92.18)	3.3 (92.23)	50.0 (92.16)
		2 $\rightarrow$ 9	2	16.7 (92.15)	43.3 (92.20)	16.7 (92.17)	56.7 (92.18)	13.3 (92.16)	73.3 (92.18)
		2 $\rightarrow$ 9	5	20.0 (92.17)	36.7 (92.17)	40.0 (92.21)	80.0 (92.16)	20.0 (92.19)	83.3 (92.15)
		2 $\rightarrow$ 9	10	20.0 (92.15)	33.3 (92.14)	50.0 (92.18)	83.3 (92.11)	53.3 (92.18)	83.3 (92.14)
		1 $\rightarrow$ 6	2	6.7 (92.18)	56.7 (92.19)	13.3 (92.18)	90.0 (92.17)	13.3 (92.19)	90.0 (92.20)
		3 $\rightarrow$ 8	2	10.0 (92.21)	66.7 (92.19)	26.7 (92.19)	60.0 (92.18)	30.0 (92.19)	63.3 (92.17)
		5 $\rightarrow$ 7	2	0.0 (92.22)	23.3 (92.21)	0.0 (92.18)	50.0 (92.19)	0.0 (92.22)	53.3 (92.16)
Tiny ImageNet	ResNet18BN	4 $\rightarrow$ 0	2	13.3 (92.21)	20.0 (92.20)	16.7 (92.20)	50.0 (92.20)	16.7 (92.21)	50.0 (92.19)
		n01629819 $\rightarrow$ n01443537	2	0.0 (49.18)	12.5 (49.18)	0.0 (49.41)	6.3 (49.41)	0.0 (49.50)	0.0 (49.50)